Predictive Multitask Learning for Streaming Code-Switching

Project Update 1
Streaming Data & Causal Baselines

Team 10

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Objective

Prefix-only code-switch prediction

- Predict whether token t+1 switches language
- Use only the utterance prefix up to token t
- Predict the duration of the upcoming language segment
- Simulates real-time bilingual text prediction

Language Coverage

All 15 language pairs

7 non-Latin pairs

7 Latin-script pairs

German–French

975k sentences

Covers 7 writing systems and 975k sentences. Language IDs come from script rules, word frequencies and context.



Goal: evaluate whether the same baseline behaves consistently across diverse language pairs.

Data Engineering Pipeline

From raw conversations to streaming prediction targets

SwitchLingua

234,172 total rows

15 language pairs

Conversation Parsing

974,994 sentences

Parsed generated conversations into individual text samples

Script-aware Tokenization

One token per word or Chinese character

Punctuation is context, **not a token**

Language ID Mapping

Script, frequency, context

Generate y_sw and y_dur

चाहिए → right

Hindi → English

y_sw = 1 y_dur = 0

Streaming Setup & Baseline Results

Prefix-only baseline evaluation

At token t, the model sees only the current prefix and predicts the event at t+1

0.555
Switch F1 (macro)

0.528
PR-AUC

0.531
Duration macro-F1

0.000
Always No-Switch F1

Streaming Example

... 敢 → 相 → 信 → she's → so → young

Input ends at t ↑

Target at t:

y_sw = 1

y_dur = 1

Future token is used only to create the target, not as input.

Logistic regression on prefix-only features, macro average over 16 strata.
Precision (0.550) and recall (0.571) are balanced.

Pair-Specific Results & Next Steps

Switch F1: best and worst pairs
Korean–English: **0.717**
Japanese–English: **0.633**
Cantonese–English: **0.409**
Filipino–English: **0.397**

Macro Mean F1: 0.555
F1 Standard Deviation: 0.076

Challenges

- Latin-script pairs rely on word frequencies; hand validation is pending
- Romanized Hindi (56% of Hindi rows) needs its own word list
- In Chinese-character text, 20-30% of positions fall inside one subword

Next Steps

- Add causal attention masking
- Build switch and duration prediction heads
- Validate Language IDs by hand
- Start multitask transformer training

Korean and Japanese are easiest, Filipino and Cantonese hardest. This is the bar the transformer models must beat.

Thank you!

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